UNIST-UTA Global Research Internship Program in Neuroimaging

Description:

Dr. Younggeun Kim (Department of Mathematical Sciences, UNIST) and Dr. Xi Zhu (Department of Bioengineering, University of Texas at Arlington (UTA)) are seeking up to TWO highly motivated students. The selected candidates will apply deep representation learning methods to ENIGMA project data (https://enigma.ini.usc.edu/) to investigate and identify complex patterns in multi-modal neuroimaging associated with psychiatric disorders.

Program Summary:

This is a 1-year program that includes a global internship at UTA, and can be extended in subsequent years depending on the students' progress. It is designed to train students to become long-term researchers in neuroimaging utilizing deep learning. Regular international Zoom meetings will be held between Drs. Kim, Zhu, and the students. During the summer or winter breaks, **students will be hired as Research Assistants at UTA** at a rate of $20 per hour (approx. $1,500 per 4 weeks), supported by a grant from the National Institutes of Health (NIH) in the US.

Students will first learn the basics of single-modality neuroimaging data analysis using deep learning. Under the joint guidance of Drs. Kim and Zhu, they will then develop conditional generative models for multi-modal data and apply these methods to analyze cluster structures in functional and structural neuroimaging data. The initial datasets will be pre-processed by Dr. Zhu's lab and made ready for deep learning applications (potentially including sources other than ENIGMA). Our goal is to publish one paper per student and have them present their work at conferences in South Korea or the US.

Eligibility and Minimum Qualification:

Open to students who are available to participate in the global internship period. **Applicants are responsible for verifying all administrative requirements (e.g., visa eligibility) in advance.**

This program is open to both undergraduate and graduate students, but the required minimum qualifications include:

a) The ability to implement vanilla autoencoders for benchmark 2D image data in GPU environments.

b) A basic understanding of functional and structural neuroimaging.

We may conduct interviews to assess these technical skills and English proficiency.

Required Documents:

Please send email to Dr. Kim (younggeun91@unist.ac.kr) along with CV and academic transcripts.